The Formal Logic of Modulign

A Complete Formal Treatment of the Modulign Standard v3.0 Architecture

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Preamble

> *This document does not explain Modulign. It proves it. Every claim the Standard makes about its own properties — validity, uniqueness, intersubjectivity, non-accidentality, chain-of-custody integrity, and scale-sensitivity — is here rendered in formal notation, derived from stated axioms, and subjected to structural verification. There are no hidden assumptions. There are no informal bridges. There are no claims that rest on good intentions rather than structural necessity.*

PART I

Foundational Definitions

1. Observable Reality and the Address Space

Definition 1.1 — Observable Phenomenon

Let Ω be the set of all observable phenomena. An element $\omega \in \Omega$ is any event, state, object, or process that:

- (a) exists in physical space-time at some coordinate (x, y, z, t),
- (b) causally produces detectable signals accessible to an observer O, and
- (c) has properties that are, in principle, intersubjectively verifiable by any competent observer with access to those signals.

Boundary condition (§30): Abstract mathematical truths, first-person conscious experience, and purely relational abstract entities are not elements of Ω. No element of Ω lacks a spatio-temporal coordinate.

Definition 1.2 — The Modulign Address Space

Let A be the Modulign address space, structured as the following tuple:

ADDRESS TUPLE

```
A = ⟨NS, DOM, SUB, REALM, G1, G2, G3, G4, NOD,
     SCL, MED, OBS, CAM, ACT, TIME, JUR, META, DYN, BG⟩

REQUIRED:  NS · DOM · SUB · REALM/G1/G2/G3/G4 · NOD
OPTIONAL:  SCL · MED · OBS · CAM · ACT · TIME · JUR · META · DYN · BG
```

Segment	Domain / Cardinality
NS	{'MGN·'} — singleton
DOM	{URB,NAT,TRN,COM,RES,EVT,SEC,INF,UNK} — |9|
SUB	D(DOM) — domain-dependent, |6–12| per domain
REALM	{SR,AT,DQ,OR,VR} — |5|
G1–G4	GeoRegistry ∪ {UNK} — hierarchical
NOD	Base36⁴ — cardinality 36⁴ = 1,679,616 per locus
SCL	{QNT,MLC,BIO,HMN,GEO,STL,CSM} — |7|
MED	{VID,AUD,SAT,SEN,TXT,IMG,MDL,BIO,DAT,MDX,SYN} — |11|
ACT	{I,T,N,P,C,E,X} — |7|
META	☐({EVID,VFYD,PROV,HOT,ALRT,CONT,NEW}) — power set
DYN	{STB,CYC,DGR,ESC,ERR,CRT} ∪ {∅ }

Definition 1.3 — The Classification Function

Let C: Ω → A be the classification function, mapping observable phenomena to Modulign addresses.

VALIDITY CONDITION

```
Valid(C(ω))  ↔  CDP(ω) → C(ω)

A classification produced outside the CDP is approximate.
It must be labeled META ∋ PROV.
```

Definition 1.4 — The Global Stream Identifier and Observational Ledger

Let GSI: Ω → UUID-v4 be the permanent identity function, immutable after first assignment:

GSI PERMANENCE AXIOM

$$\forall \omega \in \Omega,\ \forall t_1 < t_2 :\ GSI(\omega, t_1) = GSI(\omega, t_2)$$

Observational Ledger:

$$L(GSI(\omega)) = \langle (C_1, t_1), (C_2, t_2), \ldots, (C_{\square}, t_{\square}) \rangle \quad \text{where} \quad t_1 < t_2 < \ldots < t_{\square}$$

Permanence: $\forall i : (C_i, t_i) \in L(GSI(\omega))$ is immutable after emission.
Entries APPENDED. Never DELETED.

PART II

The Classification Decision Protocol

2. The CDP — Formal Procedure

The Classification Decision Protocol is a computable function:

CDP FUNCTION

$$CDP : \Omega \times ObserverState \to A \times [0,1] \times ReasoningRecord$$

$$CDP(\omega, O) = Step_8(Step_7(Step_6(Step_5(Step_4(Step_3(Step_2(Step_1(\omega, O))))))))$$

Each $Step_{\square}$ takes the output of $Step_{\square-1}$ and returns a partially-completed address with updated confidence.

Step 1 — Scene Parsing

```
SP(ω) → F(ω) ⊆ FeatureSpace
```

Extracts observable feature set F(ω) = {domain_indicators, agent_set, motion_patterns, dominant_elements, geo_cues, linguistic_cues}.

Validity condition: F(ω) ≠ ∅ . A classification on an empty feature set is not valid CDP output.

Step 2 — Dominance Resolution

```
MT(ω, d) = 1  iff  domain d accounts for >50% of semantically
significant features in ω
DOM(ω) = argmax_{d} MT(ω, d)
```

Priority ordering applied before MT:

PRIORITY FUNCTION P(DOM) → ℕ

```
P(EVT) = 1   P(SEC) = 2   P(TRN) = 3
P(COM) = 4   P(NAT) = 5   P(designed purpose) = 6   P(UNK) = 7

DR applies argmin_P before MT.
If no single domain satisfies MT: Background Domain annotation (+BG).
If genuinely unclassifiable: DOM = UNK, SUB = MIX.
```

Step 3 — Locus Resolution

LOCUS RESOLUTION + MONOTONICITY

```
LR(ω) = (REALM, G1, G2, G3, G4, NOD)

Gᵢ(ω) = geo_code_i  if geo_evidence_i ∈ F(ω),  else UNK

MONOTONICITY (Invariant I5):
Gᵢ(ω) = UNK  →  ∀j > i : Gⱼ(ω) = UNK

Cannot specify city (G4) without country (G2).
NOD ∈ ℛ_obs ∪ {0000}
```

Step 4 — Scale Assignment

Scale tiers form a total order:

```
QNT < MLC < BIO < HMN < GEO < STL < CSM
```

SCALE TRUTH CONDITION (INVARIANT I4)

```
scale(T(C)) ≠ SCL(C)  →  C is FALSE

regardless of correctness of all other segments.
Wrong scale is a category error, not an imprecision.
```

Step 5 — Medium Assignment

SYNTHETIC CONTENT BICONDITIONAL (INVARIANT I6)

```
MED = SYN  →  REALM = VR          (necessary)
REALM = VR ∧ MED = SYN  ↔  SyntheticSignature = TRUE

VR/ + :SYN is the definitive machine-readable synthetic signature.
No parsing of the full code required for synthetic detection.
```

Step 6 — Suffix Assembly and Activity Decision Tree

ACTIVITY STATE — PRIORITY DECISION TREE

```
activity(ω) =
  X  if anomaly_detected(F(ω))           [priority 1]
  E  if bounded_non_routine_event(F(ω))  [priority 2]
  T  if dominant_transit(F(ω))           [priority 3]
  N  if dominant_interaction(F(ω))       [priority 4]
  P  if dominant_production(F(ω))        [priority 5]
  C  if dominant_consumption(F(ω))       [priority 6]
  I  otherwise                           [default]

Exactly one value assigned. Priority-ordered, not multi-label.
```

^EVID OBSERVER ELIGIBILITY

```
eligible_EVID(O, DOM) ↔
  (O.type = HUM ∧ O.competence(DOM) = :A)  ∨
  (O.type = HUM ∧ O.competence(DOM) = :E ∧ consensus ≥ 2)  ∨
  (O.type = AUT ∧ AUT_certified(O, DOM, :E))  ∨
  (O.type = AUT ∧ AUT_certified(O, DOM, :A))  ∨
  (O.type = HYB ∧ human_component.competence(DOM) ≥ :E)

¬eligible_EVID(O, DOM)  →  META ∋ PROV  (Invariant I8)
```

Step 7 — Conflict Elimination

```
CONFLICT RULES — NECESSARY CONDITIONS FOR VALID
R1 (Realm-Domain):   REALM = VR → MED = SYN
                     REALM = OR → DOM ∈ {NAT,INF,COM,SEC}
                     REALM = DQ → DOM ∈ {NAT,COM,INF}

R2 (No redundancy):  ∀s_i,s_j ∈ a, i≠j : semantic(s_i) ≠ semantic(s_j)

R3 (Act-Continuum):  ACT=I → DYN ∉ {ESC,CRT}
                     ACT=X → META ∋ ALRT

R4 (Locus monoton.): G_i=UNK → ∀j>i : G_j=UNK

R5 (EVID+ISO):       META ∋ EVID → TIME ∈ ISO8601  [v3.1]

CE = INVALID if any R□ violated → ^PROV or resolution required
```

Step 8 — Confidence Scoring and Emission

```
WEIGHTED CONFIDENCE SCORE
κ = Σ_i (w_i × κ_i) / Σ_i w_i

Segment weights:
  DOM:  0.25   SUB:  0.15   LOCUS: 0.25
  NOD:  0.10   SCL:  0.10   OBS:   0.10   TIME: 0.05

κ ≥ 0.95  →  Accept. No review.
0.80 ≤ κ < 0.95  →  Accept. Periodic review.
0.60 ≤ κ < 0.80  →  ^CONT. Human review 24hrs.
κ < 0.60  →  ^PROV mandatory. No publication.
```

PART III

Trust Architecture — Formal Properties

3. Cryptographic Integrity

Theorem 3.1 — Tamper Detection

CONTENT-ADDRESSED INTEGRITY

```
H: RegistryEntry → {0,1}²⁵⁶   (SHA-256)

Integrity invariant:  stored_hash(e) = H(content(e))

TAMPER DETECTION THEOREM:
content(e) ≠ content_at_registration(e)
  →  stored_hash(e) ≠ H(content(e))
  →  tampered(e) = TRUE

P(undetected collision) ≈ 2⁻²⁵⁶ ≈ 0
Integrity is computationally guaranteed, not procedurally hoped.
```

Theorem 3.2 — Append-Only Ledger Properties

APPEND-ONLY AXIOM (INVARIANT I7)

```
∀(C,t,σ) ∈ L(g) : ¬∃ operation DELETE
  such that (C,t,σ) ∉ L(g) after operation completes.

Only permitted write:  APPEND(L(g), (C□, t□, σ□))  where  t□ > t□₋₁

ChainOfCustody(ω) ≡ L(GSI(ω))

This is a constitutive identity, not a representation.
The ledger does not document the chain of custody.
It IS the chain of custody.
```

Theorem 3.3 — Cryptographic Signature Non-Repudiation

SIGNATURE CHAIN

```
σᵢ = Sign(Cᵢ, tᵢ, σᵢ₋₁, sk)  [threshold HSM signing]

NON-REPUDIATION:
Verify(Cᵢ, tᵢ, σᵢ₋₁, σᵢ, pk) = TRUE
  →  holder of sk produced entry i at time tᵢ

CHAIN INTEGRITY:
Verify(Cᵢ, tᵢ, σᵢ₋₁, σᵢ, pk) = FALSE
  →  ledger tampered at position i or later

Threshold (k-of-n): no single party signs unilaterally.
```

PART IV

The ATNA Proof — Gettier Dissolution

4. Address-Theoretic Non-Accidentality

Definition 4.1 — Constitutive World-Engagement

Let J(C, ω) be the justificatory basis function — the set of observable features used as justificatory input in producing C.

```
CONSTITUTIVE WORLD-ENGAGEMENT
∀ Step□ ∈ CDP :  justificatory_input(Step□) ⊆ F(ω)

No step may use justificatory input not derived from F(ω).
Any step introducing independent assumptions violates this condition.
```

Definition 4.2 — The Gettier Gap and Its Closure

```
TRADITIONAL JTB — THE GAP
J(S,P) and T(P) are LOGICALLY INDEPENDENT.
J can be satisfied while T fails.
They can coincide accidentally.
→ Gettier cases are possible.
```

```
ATK — THE CLOSURE
J(C,ω) ⊆ F(ω)  and  F(ω) is causally produced by T(C).

The causal chain:
T(C)  →  physical_signals  →  F(ω)  →  CDP steps  →  J(C,ω)  →  C

There is no independence. The gap does not exist in the architecture.
```

Theorem 4.1 — ATNA (The Core Proof)

ADDRESS-THEORETIC NON-ACCIDENTALITY PRINCIPLE

```
CLAIM: A classification C produced by the full CDP is non-accidentally
true.

PROOF:
(1) CDP(ω,O) → every Step□ uses only input from F(ω).      [Def. 4.1]
(2) F(ω) is causally produced by physical properties of ω,
    i.e., by T(C) if C is true.
(3) Therefore:  J(C,ω) ⊆ F(ω) ← T(C)              [causal chain]
(4) A Gettier case requires: ∃w' where J(C,ω) is satisfied ∧ T(C) fails.
(5) In w', T(C) fails → causal chain T(C)→F(ω) is severed.
    → F(ω) lacks the features that drove CDP.
    → J(C,ω) is not satisfied.
(6) ∴ ¬∃w' such that J(C,ω) satisfied ∧ T(C) fails.
(7) ∴ C is non-accidentally true when it is true.              □
```

Corollary 4.1 — Zagzebski's Recipe Fails

WHY THE RECIPE CANNOT APPLY TO CDP

```
ZAGZEBSKI RECIPE: Take K =df true belief + X.
  Since X is fallible, construct w' where X satisfied, belief false.
  Introduce coincidence making belief true independently of X.

RECIPE ENTRY POINT: a world where X is satisfied and belief is false.

Under ATK: X = CDP.
CDP is NOT satisfiable independently of T(C).
CDP requires F(ω) as input.
F(ω) is causally produced by T(C).
If T(C) fails, F(ω) is severed, CDP cannot complete.

The recipe's entry point does not exist.  □
```

Theorem 4.2 — Intersubjectivity

ANY TWO COMPETENT OBSERVERS MUST CONVERGE

```
CLAIM: CDP(ω,O₁) = CDP(ω,O₂) for O₁,O₂ with equivalent
  competence and access to same F(ω), modulo threshold boundaries.

PROOF:
(1) MT is a >50% discrete threshold — not continuous.
    Same F(ω) → same majority determination.
```

```
(2) Activity tree is priority-ordered and discrete.
(3) Locus uses discrete geo codes.
(4) Scale uses discrete tiers.
(5) Divergence only possible at genuine threshold boundaries
    → both observers assign lower confidence (correct behavior).
    → divergence is a first-class data point, not an error.    □
```

Theorem 4.3 — Scale-Sensitivity and the Fake Barns Case

SCALE TRUTH CONDITION — FULL PROOF

```
CLAIM: SCL(C) ≠ scale(T(C))  →  C is FALSE.

PROOF:
(1) T(C) is constituted at a specific physical scale.
(2) If SCL(C) > scale(T(C)): features constituting T(C) are below
    resolution threshold of the classification. C cannot be true of T(C).
(3) If SCL(C) < scale(T(C)): truth-making properties are not constituted
    at that scale. Misidentifies the level of physical constitution.
(4) Therefore: SCL(C) ≠ scale(T(C))  →  C is FALSE.          □

APPLICATION — FAKE BARNS:
Henry's barn identification: SCL = |HMN| (human scale).
Truth-making conditions of barn-knowledge in façade-dense region:
  SCL must be |GEO| (regional scale).
SCL(C) ≠ scale(T(C))  →  Henry's classification is not knowledge-grade.
ATK dissolves fake barns by identifying the scale error.       □
```

PART V

System Invariants — What Must Always Be True

5. The Ten Invariants

The following invariants must hold at every state of the Modulign registry. Any implementation that violates any invariant is non-compliant.

Invariant	Formal Statement and Violation Consequence
I1 — GSI Permanence	∀ ω, ∀ t : GSI(ω) immutable after first assignment. Violation: entire Observational Ledger retroactively invalidated.

I2 — Ledger Monotonicity	$\forall$ L(g) : $t_1 < t_2 < \ldots < t_n$ (strictly increasing). Violation: signature chain verification fails.
I3 — ^PROV Exclusion	META $\ni$ PROV → C $\notin$ legal proceedings until human review removes PROV. Violation: inadmissible evidence admitted.
I4 — Scale Truth	SCL(C) = scale(T(C)) is necessary for C to be true. Violation: classification false regardless of other segments.
I5 — Locus Monotonicity	G_i = UNK → $\forall$ j>i : G_j = UNK. Violation: unparseable locus chain.
I6 — Synthetic Biconditional	REALM=VR $\wedge$ MED=SYN ↔ SyntheticContent(ω). Violation: false positive or false negative on synthetic detection — legally catastrophic.
I7 — EVID Append-Only	$\forall$ (C,t,σ) $\in$ L^EVID(g) : ¬DELETE. Violation: chain of custody broken; all subsequent records suspect.
I8 — Observer Competence Gate	META $\ni$ EVID → eligible_EVID(O, DOM) = TRUE. Violation: unauthorized ^EVID; retroactive ^PROV.
I9 — CDP Completeness	Valid(C) → CDP produced C (all eight steps). Violation: approximate classification must be ^PROV.
I10 — Separator Uniqueness	Each separator has exactly one syntactic role, no ambiguity. Violation: code unparseable by standard text processors.

PART VI

The Evidentiary Completeness Theorem

6. Legal Admissibility — Formal Proof

Let E_legal = {Authentication, ChainOfCustody, BestEvidence, ExpertFoundation} be the set of foundational legal admissibility requirements.

COMPLETENESS THEOREM 6.1

```
CLAIM: ∀e ∈ E_legal : satisfied(e, C^EVID) = TRUE
  where C^EVID is a classification bearing META ∋ EVID
  produced by the full CDP.

PROOF BY ELEMENT:

AUTHENTICATION (FRE 901(b)(9)):
  CDP is a documented, reproducible, 8-step process.
  JSON record includes full reasoning record ρ.
  → satisfied(Authentication, C^EVID) = TRUE.       □
```

```
CHAIN OF CUSTODY:
  L(GSI(ω)) with σᵢ = Sign(Cᵢ, tᵢ, σᵢ₋₁, sk) constitutes CoC.
  ChainOfCustody(ω) ≡ L(GSI(ω))  [Def. 3.2]
  No separate CoC documentation required.
  → satisfied(ChainOfCustody, C^EVID) = TRUE.       □

BEST EVIDENCE (FRE 1001-1008):
  'Original' = GSI-ID + Observational Ledger.
  GSI-ID: UUID-v4, P(collision) ≈ 2⁻¹²², permanent, content-addressed.
  Duplicates detectable via hash verification.
  → satisfied(BestEvidence, C^EVID) = TRUE.         □

EXPERT FOUNDATION (FRE 702/Daubert):
  Sufficient facts:  observable feature set F(ω).
  Reliable method:   CDP (documented, reproducible, known error rate).
  Reliable application: %OBS observer annotation.
  → satisfied(ExpertFoundation, C^EVID) = TRUE.     □

∴ ∀e ∈ E_legal : satisfied(e, C^EVID) = TRUE.      □
```

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The address system for what happened is formally sound.